

Textbook-style Student Notes
BCA30040 – INTRODUCTION TO CLOUD COMPUTING
UNIT I – UNDERSTANDING CLOUD COMPUTING

Primary Reference: Thomas Erl, Ricardo Puttini & Zaigham Mahmood, *Cloud Computing: Concepts, Technology & Architecture*

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Syllabus: 6 Lectures

1. Introduction to Cloud Computing
2. Characteristics of Cloud Computing
3. Evolution of Cloud Computing
4. Cloud Computing Use Cases
5. Benefits and Challenges of Cloud Computing

The syllabus specifically lists these five topics under Unit I.

Unit-I → Reference Book Mapping

Unit-I Syllabus Topic	Reference Book Chapter	Page Number
1. Introduction to Cloud Computing	Chapter 3 – Understanding Cloud Computing	25–40
2. Characteristics of Cloud Computing	Chapter 4 – Fundamental Concepts and Models	58–62
3. Evolution of Cloud Computing	Chapter 3 – Understanding Cloud Computing	26–32
4. Cloud Computing Use Cases	Chapter 3 – Understanding Cloud Computing; Chapter 4 – Fundamental Concepts and Models	28–40; 58–62
5. Benefits of Cloud Computing	Chapter 3 – Understanding Cloud Computing	40–44
6. Challenges of Cloud Computing	Chapter 3 – Understanding Cloud Computing	45–48

1. Introduction to Cloud Computing

1.1 What is Cloud Computing?

Cloud computing is a computing approach in which **IT resources and services are made available remotely through a cloud environment**.

Instead of an organization purchasing, installing and maintaining all the required hardware and software itself, it can obtain computing capabilities from a **cloud provider** when required.

The important idea in the reference book is that computing capabilities can be **consumed as services**. The focus is therefore not only on physical computers but on accessing and provisioning IT resources remotely.

Simple meaning

Cloud Computing = Using computing resources as services whenever required instead of owning and managing everything yourself.

Example

Suppose a college wants to conduct online examinations.

Traditional approach

The college may need to:

- purchase servers,
- install operating systems,
- install database software,
- maintain storage,
- maintain networking,
- hire people to manage the infrastructure.

Cloud approach

The college can obtain computing resources from a cloud provider and use them for:

- examination application,
- database,
- student data,
- storage,
- backup.

After the examination period, the college can reduce or release resources.

Main idea: The college consumes computing capability when required rather than permanently owning all the infrastructure.

1.2 Cloud Provider and Cloud Consumer

The reference book emphasizes two important roles:

Cloud Provider

A **cloud provider** is the organization that makes cloud-based IT resources and services available.

Examples:

- Amazon Web Services
- Microsoft Azure
- Google Cloud

Cloud Consumer

A **cloud consumer** is the person or organization that uses the cloud-based resource or service.

For example:

A college uses cloud storage to store student documents.

- Cloud provider → provides storage
- College → consumes storage

The provider–consumer relationship is an important part of the book's conceptual model.

Easy example

Think about electricity.

You normally do not build your own power plant.

Instead:

Electricity provider → supplies electricity → Consumer

Similarly:

Cloud provider → supplies computing resources → Cloud consumer

This is why cloud computing is often associated with **service consumption**.

1.3 What can be obtained from the Cloud?

Cloud environments can provide different types of IT capabilities, such as:

- Computing power
- Storage
- Databases
- Networking
- Application platforms
- Software
- Backup facilities
- Application hosting

For example, an organization developing a mobile application may need:

Compute + Database + Storage + Networking

Instead of purchasing all these resources, it can obtain them from a cloud environment.

The uploaded reference material describes cloud environments as pools of IT resources that can be leased and used according to requirements.

1.4 Cloud Computing vs Traditional Computing

Traditional Computing	Cloud Computing
Organization purchases infrastructure	Organization obtains resources from provider
Large initial investment	Lower initial investment possible
Organization manages infrastructure	Provider manages much of the infrastructure
Capacity may remain unused	Resources can be adjusted according to demand
Scaling may require purchasing hardware	Resources can be scaled more easily
Maintenance is organization's responsibility	Much infrastructure maintenance is handled by provider

Example

A startup wants to launch an online shopping website.

With traditional computing, it may need to purchase servers before knowing how many customers it will get.

With cloud computing, it can start with fewer resources and increase resources if the number of customers increases.

2. Characteristics of Cloud Computing

According to the uploaded reference book, **Chapter 4 identifies six important cloud characteristics:**

1. On-demand usage
2. Ubiquitous access
3. Multitenancy and resource pooling
4. Elasticity
5. Measured usage
6. Resiliency

Let us understand each one.

2.1 On-Demand Usage

Meaning

On-demand usage means that cloud consumers can obtain and use IT resources **when they are required**.

The consumer can often self-provision resources without having to wait for someone to physically install hardware.

The reference material describes self-provisioning and automated access to cloud resources as important aspects of on-demand usage.

Example

Suppose an online shopping company expects a large sale tomorrow.

It needs additional servers.

Instead of purchasing a new physical server, it can provision additional cloud computing resources when required.

After the sale, unnecessary resources can be released.

Remember

On-demand = Use when needed.

2.2 Ubiquitous Access

Meaning

Ubiquitous access means that cloud services are designed to be accessible from different locations and through different types of devices, provided suitable connectivity and access mechanisms are available.

The reference material emphasizes broad accessibility through different devices, interfaces and technologies.

Example

A student saves a document in cloud storage.

The student can access the document using:

- Laptop
- Desktop
- Tablet
- Smartphone

from different locations, provided the necessary network access and authentication are available.

Remember

Ubiquitous access = Access from many places/devices.

2.3 Multitenancy and Resource Pooling

This concept has two related parts.

A. Multitenancy

Multitenancy means that a cloud environment can serve multiple customers or **tenants** using shared underlying resources while maintaining logical separation between them.

Example

Suppose 1,000 students use the same cloud-based learning management system.

The system infrastructure is shared, but:

- Student A sees Student A's information.
- Student B sees Student B's information.

Their data is logically separated.

Important point

Shared infrastructure does not mean that users automatically see each other's data.

B. Resource Pooling

Resource pooling means that physical and virtual resources can be dynamically assigned and reassigned to different consumers according to requirements.

Virtualization is commonly used to support resource pooling.

Example

A cloud provider may have a large pool of servers.

At one moment:

- Application A needs more CPU.
- Application B needs more memory.
- Application C needs less computing power.

Cloud resources can be dynamically allocated according to demand.

Remember

Multitenancy = Many consumers share the environment.

Resource pooling = Resources are dynamically allocated from a common pool.

2.4 Elasticity

Meaning

Elasticity is the ability of cloud resources to **increase or decrease according to workload demand**.

When demand increases:

Resources ↑

When demand decreases:

Resources ↓

The reference material specifically describes elasticity as adapting IT resources to changing workloads.

Example

Consider an online ticket-booking website.

Normally:

1,000 users → 2 servers

During a popular movie release:

20,000 users → additional servers required

After the rush:

1,000 users → extra servers can be reduced

This ability to adjust resources is elasticity.

Remember

Elasticity = Increase + Decrease according to demand.

2.5 Measured Usage

Meaning

Cloud environments can **measure how much of a resource or service is being used.**

Measurement can support:

- monitoring,
- management,
- usage analysis,
- usage-based charging.

The reference material explicitly connects measured usage with monitoring, management and charging.

Example

Suppose a company uses:

- 100 GB storage
- 500 GB data transfer
- 20 hours of computing

The cloud environment can measure this usage.

The organization can therefore understand:

How much cloud resource did we actually consume?

Remember

Measured usage = Measure what you consume.

2.6 Resiliency

Meaning

Resiliency means the ability of a cloud environment to continue providing services or recover from failures.

Cloud environments can use techniques such as:

- redundancy,
- replication,
- failover,
- distributed resources.

Example

Suppose a cloud server fails.

If another server is available to take over the workload, the application can continue operating with minimal interruption.

The reference book connects high availability and reliability with cloud architectures and failover support.

Remember

Resiliency = Continue or recover when something fails.

2.7 Six Characteristics – Quick Revision

Characteristic	Simple Meaning	Example
On-demand usage	Use resources when required	Create a server when needed
Ubiquitous access	Access from different locations/devices	Access cloud files from phone
Multitenancy	Multiple consumers share environment	Many organizations use same cloud service
Resource pooling	Resources allocated from common pool	CPU assigned according to demand
Elasticity	Increase/decrease resources	Add servers during sale
Measured usage	Usage is measured	Measure storage used

Characteristic	Simple Meaning	Example
Resiliency	Continue/recover after failure	Backup server takes over

Exam tip: The book groups **multitenancy and resource pooling** together, so students should remember them as one characteristic.

3. Evolution of Cloud Computing

Cloud computing did not appear suddenly. It developed through several earlier computing and networking approaches.

The reference book's Chapter 3 discusses the **history of cloud computing, business drivers and technology innovations**.

A simplified learning sequence is:

Traditional Computing

↓

Networked / Distributed Computing

↓

Virtualization and Data Centers

↓

Web-based and Service-Oriented Computing

↓

Cloud Computing

This teaching sequence is also reflected in the uploaded Unit-I material.

3.1 Traditional / Centralized Computing

In early computing environments, computing was generally centered around dedicated computers or centralized systems.

Organizations owned and managed their computing equipment.

Example

A college purchases a physical server and keeps it inside its own server room.

The college is responsible for:

- hardware,
- operating system,
- maintenance,
- backup,

- security.
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3.2 Networked Computing

As computer networks developed, computers could communicate and share information.

Computing was no longer restricted to a single machine.

Example

Computers in different departments of a college communicate through a campus network.

3.3 Distributed Computing

In distributed computing, processing can be performed across multiple connected computers.

Instead of depending completely on one computer, workloads can be distributed among multiple systems.

Example

A large scientific calculation can be divided among multiple computers.

3.4 Virtualization

Virtualization became an important enabling technology.

A physical server can be divided into multiple **virtual machines**, allowing multiple computing environments to run on the same physical hardware.

Example

One physical server:

Physical Server

→ Virtual Machine 1

→ Virtual Machine 2

→ Virtual Machine 3

Each virtual machine can run its own operating system and applications.

This improves utilization of physical resources.

3.5 Web-Based Services

The growth of the Internet and Web made it possible to access applications and information remotely.

Users increasingly began accessing services through web browsers instead of installing everything locally.

Example

Instead of installing an application on every computer, users access a web-based application through a browser.

3.6 Service-Oriented Computing

The development of service-oriented approaches encouraged organizations to think about IT capabilities as **reusable services**.

The reference book explains that the movement toward service orientation and reusable components helped support more flexible and service-oriented approaches to computing.

Example

Instead of building one large application containing everything, an organization may use separate services for:

- authentication,
 - payment,
 - storage,
 - customer management.
-

3.7 Cloud Computing

Cloud computing brought many of these developments together.

It combines concepts such as:

- network access,
- distributed resources,
- virtualization,
- data centers,
- web-based access,
- service-oriented computing,

- resource sharing,
- scalable resource management.

Why did cloud computing emerge?

Important drivers included:

- increasing data volumes,
- increasing computing requirements,
- need for scalability,
- need for flexible IT resources,
- need for remote access,
- business interest in consuming IT as services.

These drivers are reflected in the uploaded textbook-aligned material.

3.8 Evolution – Easy Example

Imagine a college over time:

Stage 1

College owns one computer.



Stage 2

College has many networked computers.



Stage 3

College uses several servers.



Stage 4

Virtualization allows multiple virtual servers on physical servers.



Stage 5

College starts using web-based applications.



Stage 6

College obtains computing resources from a cloud provider.

This progression helps students understand **why cloud computing developed**.

4. Cloud Computing Use Cases

A **use case** describes a practical situation in which cloud computing is used to solve a problem or provide a service.

The important question is:

Where can cloud computing actually be used?

Common use cases include:

4.1 Application Hosting

Organizations can host applications in cloud environments instead of maintaining their own physical servers.

Example

A college hosts its student management application on cloud infrastructure.

Students and staff access the application through the Internet.

4.2 Data Storage

Cloud computing can be used to store large quantities of data.

Example

Students store:

- assignments,
- presentations,
- project files,
- photographs

using cloud storage.

4.3 Backup and Disaster Recovery

Cloud resources can be used to maintain backup copies of important information.

Example

A college maintains a backup of its student database in a separate cloud environment.

If the local server fails, the backup can be used for recovery.

4.4 Development and Testing

Developers can obtain computing resources for developing and testing applications.

Example

A student project requires a temporary server for testing.

Instead of purchasing a physical server, the student team can provision a cloud resource, test the application and release the resource after testing.

4.5 Web and Mobile Applications

Cloud environments can host applications used by large numbers of users.

Example

An online food-ordering application may use cloud computing for:

- application hosting,
 - database,
 - storage,
 - networking.
-

4.6 High-Demand Applications

Cloud computing is useful when workload changes significantly.

Example

An online examination system normally has moderate usage but experiences very high usage during examination time.

Cloud resources can be scaled to accommodate the increased workload.

4.7 Business Continuity

Cloud environments can help organizations maintain access to IT resources even when local infrastructure experiences problems.

Example

If an organization's local server becomes unavailable, cloud-based backup or alternate infrastructure can support business operations.

5. Benefits of Cloud Computing

The reference book specifically identifies three major goals/benefits in Chapter 3:

1. Reduced investments and proportional costs
2. Increased scalability
3. Increased availability and reliability

Let's understand them.

5.1 Reduced Initial Investment

In traditional computing, an organization may need to purchase:

- servers,
- storage,
- networking equipment,
- software,
- data-center facilities.

Cloud computing can reduce the need for large initial infrastructure investments.

Example

A startup does not need to purchase several expensive servers before launching its website. It can obtain required computing resources from a cloud provider.

5.2 Proportional / Usage-Based Costs

Cloud environments can support a **pay-for-use** approach.

The reference book explains that cloud resources can be leased and billed according to measured usage.

Example

If a company uses more storage, its usage increases.

If it reduces its storage requirement, its resource requirement can also decrease.

Simple idea

Use more → potentially pay more

Use less → potentially pay less

5.3 Increased Scalability

Scalability means the ability to accommodate increasing workloads by increasing available resources.

Example

An e-commerce website normally serves 5,000 users.

During a festival sale:

5,000 → 50,000 users

Cloud resources can be increased to handle the additional workload.

The book highlights the ability of cloud environments to accommodate unpredictable usage fluctuations.

5.4 Increased Availability

Availability means that an IT resource is accessible when users need it.

Cloud environments can use redundant and resilient infrastructure to improve availability.

Example

An online shopping website should ideally remain accessible even if one underlying component fails.

5.5 Increased Reliability

Reliability refers to the ability of an IT resource to avoid failures or recover from them.

The book explains that modular cloud architectures can provide failover support that improves reliability.

Example

If one server fails, another server may take over the workload.

6. Challenges of Cloud Computing

Cloud computing provides significant benefits, but adoption also creates challenges.

The reference book identifies four major challenges in Chapter 3:

1. Increased security vulnerabilities
 2. Reduced operational governance control
 3. Limited portability between cloud providers
 4. Multi-regional compliance and legal issues
-

6.1 Increased Security Vulnerabilities

When an organization moves data and IT resources to a cloud provider, security becomes a shared concern.

The organization has to trust the cloud environment and its security controls.

The reference book discusses expanded trust boundaries and the security implications of sharing cloud resources.

Example

A company stores confidential customer information in a public cloud.

Security now depends on:

Company security controls + Cloud provider security controls

Therefore, improper configuration or weak security controls can create risks.

Remember

Cloud does not mean automatically secure.

6.2 Reduced Operational Governance Control

In an on-premise environment, an organization generally has direct control over its infrastructure.

In a cloud environment, some operational control is transferred to the cloud provider.

The reference book notes that cloud consumers generally have less governance control over cloud-hosted infrastructure than over their own on-premise resources.

Example

A college owns its own server.

It can directly decide:

- when maintenance occurs,
- how hardware is configured,
- where the server is located.

If the server is hosted by a cloud provider, many of these operational decisions are controlled by the provider.

6.3 Limited Portability

Portability refers to how easily applications and data can be moved from one cloud environment to another.

A problem can occur when an application depends heavily on provider-specific technologies.

The reference book explains that proprietary cloud characteristics can make migration between providers difficult.

Example

Suppose a company develops an application using several special features of Cloud Provider A.

Later, it wants to move to Cloud Provider B.

If Provider B does not support those features, the company may have to modify the application.

This is commonly associated with **vendor lock-in**.

6.4 Multi-Regional Compliance and Legal Issues

Cloud providers may operate data centers in different geographical regions.

The consumer may not always control the physical location of its data.

Different countries and industries may have different rules regarding:

- data storage,
- privacy,
- security,
- access,
- disclosure.

The reference book specifically identifies geographical data location as a source of legal and regulatory concerns.

Example

Suppose a company is required by regulation to keep certain customer data within a particular country.

If its cloud provider stores that data in another country, the organization may face compliance issues.

7. Benefits vs Challenges – Quick Comparison

Benefits	Challenges
Reduced initial investment	Security concerns
Usage-based/proportional costs	Reduced operational control
Increased scalability	Limited portability
Increased availability	Legal and compliance issues
Increased reliability	Dependence on provider/network

8. Complete Unit-I Example

Consider a **college online examination system**.

The college has 20,000 students.

Traditional approach

The college purchases servers to handle the examination system.

But the servers may remain underutilized after examinations.

Cloud approach

The college obtains computing resources from a cloud provider.

During normal days:

Low resources

During examination:

More resources

After examination:

Resources reduced

This single example demonstrates:

- **On-demand usage** → resources obtained when required
- **Elasticity** → resources increase during examination
- **Measured usage** → resource consumption can be monitored
- **Resource pooling** → resources come from cloud infrastructure
- **Availability** → infrastructure can be designed for continued service
- **Reduced investment** → no need to purchase all infrastructure
- **Security challenge** → student data must be protected
- **Governance challenge** → some infrastructure control belongs to provider
- **Portability challenge** → migration may be difficult if provider-specific services are used
- **Compliance challenge** → data location and regulations must be considered

This is a very good **integrated example for examination answers**.

9. What Students Should Understand

Why did cloud computing come?

Traditional computing → networked/distributed computing → virtualization → web/service-oriented computing → cloud.

Then:

What makes cloud different?

On-demand → ubiquitous access → resource sharing → elasticity → measurement → resiliency.

Then:

Where is it used?

Hosting → storage → backup → development/testing → web/mobile applications.

Finally:

Why do organizations adopt it, and what should they worry about?

Benefits → lower investment, scalability, availability/reliability

Challenges → security, governance, portability, legal/compliance.

This structure closely follows the conceptual organization of the **Erl-Puttini-Mahmood** reference book. The book itself is deliberately vendor-neutral and organized around fundamental cloud concepts, models, mechanisms and architectures.

Reference used

Thomas Erl, Ricardo Puttini & Zaigham Mahmood, *Cloud Computing: Concepts, Technology & Architecture*

- **Chapter 3 – Understanding Cloud Computing:** history/evolution, business drivers, terminology, benefits and challenges.
- **Chapter 4 – Fundamental Concepts and Models:** cloud characteristics, roles/boundaries, delivery and deployment models.
- Your uploaded syllabus confirms these five topics constitute **Unit I – Understanding Cloud Computing (6 lectures)**.